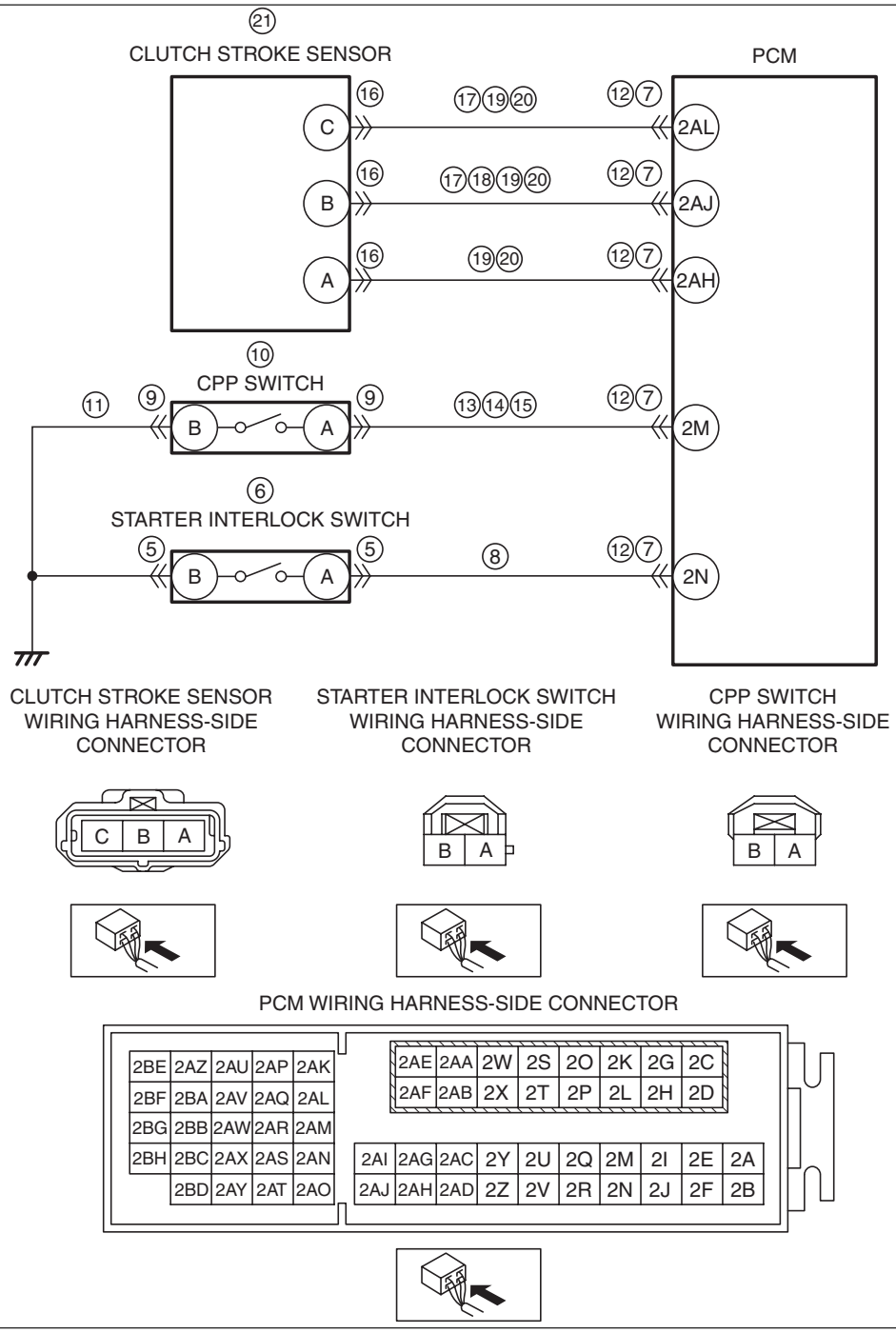
DTC P176E:00 [SKYACTIV-D 1.5]

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DTC P176E:00	Clutch stroke sensor/CPP switch/starter interlock switch correlation problem
DETECTION CONDITION	• If the clutch stroke sensor circuit voltage **above 4.86 V** or **below 0.14 V** for **5 s**, the PCM determines that the clutch stroke sensor circuit has a malfunction. • Correlation error between the clutch stroke sensor and the CPP switch, starter interlock switch is detected when any of the following conditions is met. — Correlation error content: • CPP switch is ON and clutch pedal stroke amount is **less than 1%** • CPP switch is OFF and clutch pedal stroke amount is **larger than 70%** • Starter interlock switch is ON and clutch pedal stroke amount is **less than 10%** • CPP switch is OFF and starter interlock switch is ON **MONITORING CONDITIONS** — The following DTCs are not detected: • CPP switch: P0704:00 **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	**Caution** **• Inspect the CPP switch with it installed to the clutch pedal, otherwise the CPP switch may not operate normally after inspection. If the CPP switch is removed from the clutch pedal, replace the CPP switch with a new one.** • CPP switch, clutch stroke sensor and/or starter interlock switch loose • Starter interlock switch connector or terminals malfunction • Starter interlock switch malfunction • PCM connector or terminals malfunction • Short to ground in wiring harness between starter interlock switch terminal A and PCM terminal 2N • CPP switch connector or terminals malfunction • CPP switch malfunction • Open circuit in wiring harness between CPP switch terminal B and body ground • Short to ground in wiring harness between CPP switch terminal A and PCM terminal 2M • Short to power supply in wiring harness between CPP switch terminal A and PCM terminal 2M • Open circuit in wiring harness between CPP switch terminal A and PCM terminal 2M • Clutch stroke sensor connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Clutch stroke sensor terminal C—PCM terminal 2AL — Clutch stroke sensor terminal B—PCM terminal 2AJ • Short to power supply in wiring harness between clutch stroke sensor terminal B and PCM terminal 2AJ • Clutch stroke sensor circuits are shorted to each other. • Open circuit in wiring harness between the following terminals: — Clutch stroke sensor terminal C—PCM terminal 2AL — Clutch stroke sensor terminal B—PCM terminal 2AJ — Clutch stroke sensor terminal A—PCM terminal 2AH • Clutch stroke sensor malfunction • PCM malfunction

DTC P176E: 00	Clutch stroke sensor/CPP switch/starter interlock switch correlation problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
2	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P0704:00 also present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC P0704:00 [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT INSTALLATION OF CPP SWITCH, CLUTCH STROKE SENSOR AND STARTER INTERLOCK SWITCH** • Inspect installation of CPP switch, clutch stroke sensor and starter interlock switch. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/INSTALLATION [F66M-R].) (See CLUTCH MASTER CYLINDER REMOVAL/ INSTALLATION [F66M-R].) (See STARTER INTERLOCK SWITCH REMOVAL/INSTALLATION [F66M-R].) • Are the CPP switch, clutch stroke sensor and starter interlock switch installed securely?	Yes	Go to the next step.
		No	Retighten the malfunctioning switch and/or sensor, then go to Step 22. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/ INSTALLATION [F66M-R].) (See CLUTCH MASTER CYLINDER REMOVAL/ INSTALLATION [F66M-R].) (See STARTER INTERLOCK SWITCH REMOVAL/ INSTALLATION [F66M-R].)
4	**DETERMINE IF TEMPORARY MALFUNCTION DUE TO DTC P176E:00-RELATED PART** • Inspect the following: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — PCM PID: • CPP (%) — PCM terminal 2N voltage • Is there any malfunction? (See PCM INSPECTION [SKYACTIV-D 1.5].)	Yes	If the PCM terminal 2N voltage is not normal: • Go to the next step. If the CPP PID (%) is not normal: • Go to Step 9.
		No	Intermittent concern exists. • Perform the “INTERMITTENT CONCERN TROUBLESHOOTING” procedure. (See INTERMITTENT CONCERN TROUBLESHOOTING [SKYACTIV-D 1.5].)
5	**INSPECT STARTER INTERLOCK SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the starter interlock switch connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 22.
		No	Go to the next step.
6	**INSPECT STARTER INTERLOCK SWITCH** • Inspect the starter interlock switch. (See STARTER INTERLOCK SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the starter interlock switch, then go to Step 22. (See STARTER INTERLOCK SWITCH REMOVAL/ INSTALLATION [F66M-R].)
		No	Go to the next step.
7	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 22.
		No	Go to the next step.
8	**INSPECT STARTER INTERLOCK SWITCH SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the starter interlock switch and PCM connectors are disconnected. • Inspect for continuity between starter interlock switch terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between starter interlock switch terminal A and PCM terminal 2N. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 22.
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**INSPECT CPP SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the CPP switch connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 22.
		No	Go to the next step.
10	**INSPECT CPP SWITCH** • Inspect the CPP switch. (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the CPP switch, then go to Step 22. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/ INSTALLATION [F66M-R].)
		No	Go to the next step.
11	**INSPECT CPP SWITCH GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the CPP switch connector is disconnected. • Inspect for continuity between CPP switch terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal B and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect for the following: — Open circuit between CPP switch and body ground — Loose or lifting ground point • Repair or replace the malfunctioning part. Go to Step 22.
12	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 22.
		No	Go to the next step.
13	**INSPECT CPP SWITCH SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the CPP switch and PCM connectors are disconnected. • Inspect for continuity between CPP switch terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal A and PCM terminal 2M. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 22.
		No	Go to the next step.
14	**INSPECT CPP SWITCH SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the CPP switch and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the CPP switch terminal A (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal A and PCM terminal 2M. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 22.

STEP	INSPECTION		ACTION
15	**INSPECT CPP SWITCH SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Verify that the CPP switch and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between CPP switch terminal A (wiring harness-side) and PCM terminal 2M (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal A and PCM terminal 2M. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 22.
16	**INSPECT CLUTCH STROKE SENSOR CONNECTOR CONDITION** • Disconnect the clutch stroke sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 22.
		No	Go to the next step.
17	**INSPECT CLUTCH STROKE SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the clutch stroke sensor and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Clutch stroke sensor terminal C — Clutch stroke sensor terminal B • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Clutch stroke sensor terminal C—PCM terminal 2AL • Clutch stroke sensor terminal B—PCM terminal 2AJ **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 22.
		No	Go to the next step.
18	**INSPECT CLUTCH STROKE SENSOR SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the clutch stroke sensor and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the clutch stroke sensor terminal B (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between clutch stroke sensor terminal B and PCM terminal 2AJ. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 22.

STEP	INSPECTION		ACTION
19	**INSPECT CLUTCH STROKE SENSOR CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the clutch stroke sensor and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between clutch stroke sensor terminals C, B and A (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Clutch stroke sensor terminal C—PCM terminal 2AL • Clutch stroke sensor terminal B—PCM terminal 2AJ • Clutch stroke sensor terminal A—PCM terminal 2AH **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 22.
		No	Go to the next step.
20	**INSPECT CLUTCH STROKE SENSOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the clutch stroke sensor and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Clutch stroke sensor terminal C—PCM terminal 2AL — Clutch stroke sensor terminal B—PCM terminal 2AJ — Clutch stroke sensor terminal A—PCM terminal 2AH • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Clutch stroke sensor terminal C—PCM terminal 2AL • Clutch stroke sensor terminal B—PCM terminal 2AJ • Clutch stroke sensor terminal A—PCM terminal 2AH **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 22.
21	**INSPECT CLUTCH STROKE SENSOR** • Inspect the clutch stroke sensor. (See CLUTCH STROKE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the clutch master cylinder, then go to the next step. (See CLUTCH MASTER CYLINDER REMOVAL/INSTALLATION [F66M-R].)
		No	Repeat Step 4.
22	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Depress and release the clutch pedal. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
23	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.